

Salil Patki

Naperville, IL · College Station, TX

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Objective

Mechanical Engineering student at Texas A&M University seeking a robotics engineering internship. Strong hands-on background in CAD design, rapid prototyping, CNC/waterjet manufacturing, and competitive robotics, with demonstrated leadership in taking robotic systems from concept through fabrication and testing.

Education

Texas A&M University, College Station, TX

Expected May 2030

B.S. in Mechanical Engineering

- Craig and Galen Bowen Engineering Honors Honoree
- President's Endowed Scholar Recipient (~250 of 15,000 incoming freshmen)
- National Merit Scholar (top ~0.5% nationally)

Waubonsie Valley High School, Aurora, IL

Aug 2022 – May 2026

High School Diploma

- GPA: 4.3/4.0, Summa Cum Laude
- Science Honor Society Member
- Activities: Science Olympiad, SkillsUSA

Technical Skills

- **Design & CAD:** Parametric CAD/CAM Modeling, Design for Manufacturing (DFM), Mechanical Assembly Design, Tolerancing
- **Simulation & Analysis:** Computational Fluid Dynamics (CFD), OpenFOAM (OpenCFD), Underwater/Submersible Design, Failure Analysis & Troubleshooting
- **Manufacturing & Prototyping:** 3D Printing (FDM, SLA), CNC Machining, Waterjet Cutting, 3D Printer Operation, Calibration & Repair (Bambu, Anycubic, Prusa)
- **Robotics:** ROV & Competition Robot Design, Custom Drivetrains, Actuation & Mechanism Design, Machine Vision, Iterative Prototyping
- **Programming:** Java, C++
- **Tools:** Microsoft Office Suite

Robotics & Engineering Projects

FIRST Tech Challenge – Robotics Design & Team Lead

CAD · CNC · FDM · Waterjet

- Designed a **custom omnidirectional drivetrain** for full holonomic movement, enabling precise multi-axis maneuvering during matches
- Engineered an **indexer subsystem** to reliably feed and sequence game elements, and a **shooter mechanism** tuned for consistent projectile launch velocity and accuracy
- Integrated CNC-machined, waterjet-cut, and 3D-printed components into functional, serviceable subsystems
- Iterated through multiple prototypes to improve reliability and performance across competition cycles
- Managed team finances and **raised \$15,000** to fund robotics operations and STEM education outreach
- Led a multidisciplinary team through design reviews, build cycles, and competition testing

SeaPerch Underwater ROV Competition

CAD · FDM · SLA · CFD (OpenFOAM) · Systems Design

- Ranked **4th out of 9,000+ teams worldwide** and won the **Innovation Award** for the best ROV design in competition
- Designed a custom **ballasting system** to achieve neutral buoyancy and stable depth control underwater
- Integrated a **vision system** for underwater navigation and task completion
- Engineered **venturi-effect jet propulsion** to improve thrust efficiency and maneuverability
- Applied computational fluid dynamics (CFD) in OpenFOAM to optimize hull hydrodynamics, drag reduction, and stability
- Fabricated a waterproofed, hydrodynamic ROV chassis using additive manufacturing and iterative testing

Experience

Assistant

Aug 2026 – Present

Texas A&M Corps of Cadets, College Station, TX

- Support daily administrative and operational tasks within the Corps of Cadets organization
- Collaborate with staff and cadets to coordinate logistics and ensure smooth program execution
- Maintain organization and attention to detail in a fast-paced, structured environment

Soundboard Operator

Jan 2026 – May 2026

Waubonsie Valley High School, Aurora, IL

- Operated and troubleshoot audio mixing systems for live events, applying technical problem-solving under time pressure
- Set up, calibrated, and tested electronic equipment prior to performances

Awards & Honors

- National Merit Scholar – National Merit Scholarship Program
- Presidential Scholarship – Texas A&M
- Excellence in Youth Leadership – City of Aurora
- Commendation – Illinois General Assembly
- IP Scholar – Indian Prairie School District

Languages

English (Native) · German (Full Professional) · Dutch (Full Professional) · Russian (Limited Working)